COMPUTER APPLICATION I — MODULE 1 THEORY NOTES

TVET ICT Revision Notes | Module 1 | KNEC-style preparation

Table of Contents

TOPIC 1: INTRODUCTION TO COMPUTERS 3

1.1 Definition of a computer 3

1.6 Types of computers 4

TOPIC 2: COMPUTER SYSTEMS 5

TOPIC 3: COMPUTER HARDWARE 6

3.1 Definition 6

3.2 Input devices 6

3.3 Output devices 9

3.4 Central Processing Unit (CPU) 11

3.5 Storage devices 11

3.6 Primary and secondary storage 11

TOPIC 4: COMPUTER SOFTWARE 11

4.1 Definition 11

4.2 Types of software 12

4.3 System software versus application software 12

4.4 Language translators 13

TOPIC 5: OPERATING SYSTEMS 13

5.1 Definition 13

5.4 Graphical User Interface (GUI) and Command-Line Interface (CLI) 14

5.5 Starting and shutting down a computer 14

TOPIC 6: DESKTOP AND FILE MANAGEMENT 14

6.1 Desktop 14

6.2 Files and folders 15

6.3 Common file operations 15

6.4 Common keyboard shortcuts 15

TOPIC 7: COMPUTER CARE, SAFETY AND SECURITY 16

7.1 Computer care and maintenance 16

7.2 Computer security threats 17

TOPIC 8: BASIC WORD PROCESSING 18

8.1 Definition 18
8.2 Common word-processing operations 18
8.3 Text formatting 18
TOPIC 9: BASIC SPREADSHEET APPLICATIONS 19
9.1 Definition 19
9.2 Spreadsheet terminology 19
9.3 Common spreadsheet formulas 20
TOPIC 10: BASIC DATABASE APPLICATIONS 20
10.1 Definition 20
10.2 Database terminology 20
10.4 Advantages of databases 21
TOPIC 11: BASIC PRESENTATION APPLICATIONS 21
11.1 Definition 21
11.2 Components of a presentation 21
TOPIC 12: COMPUTER NETWORKS AND THE INTERNET 22
12.1 Definition of a computer network 22
12.2 Types of networks 22
12.3 Network devices 22
12.4 Internet terminology 22
END-OF-MODULE 1 REVISION QUESTIONS 23

TOPIC 1: INTRODUCTION TO COMPUTERS

1.1 Definition of a computer

A computer is an electronic device that accepts data as input, processes it according to a set of instructions, stores it, and produces meaningful information as output.

1.2 Basic functions of a computer

1. Input

Entering data into the computer

2. Processing

Converting data into meaningful information

3. Output

Presenting the processed information

4. Storage

Saving data and information for future use

Example: In a college examination system, marks are entered, the computer calculates totals and grades, results are displayed, and the results are saved in a database.

1.3 Characteristics of computers

1. Speed: Performs operations very quickly.

2. Accuracy: Produces accurate results when the data and instructions are correct.

3. Storage capacity: Stores large amounts of data.

4. Diligence: Performs repetitive tasks without becoming tired.

5. Versatility: Performs different tasks, such as calculations, communication, and document preparation.

6. Automation: Works automatically after receiving instructions.

7. Reliability: Produces consistent results under the same conditions.

1.4 Limitations of computers

- Cannot independently apply human judgment in every situation.

- Depend on instructions and supplied data.

- May produce incorrect results because of incorrect input or instructions.

- Require electricity or a suitable power source.

- Can be affected by malware, hardware failure, and data loss.

1.5 Applications of computers

Area	Application
Education	E-learning, examination processing
Banking	ATM transactions and account management
Healthcare	Patient records and medical imaging
Business	Payroll, stock control, accounting
Communication	Email, video conferencing
Transport	Ticket booking and navigation
Government	Records management and online services
Entertainment	Gaming, music, video editing

1.6 Types of computers

According to size and processing power

- Supercomputers: Used for complex scientific calculations, climate modelling, and simulations.

- Mainframe computers: Handle large volumes of transactions and data for large organisations.

- Minicomputers: A historical category of midrange multi-user computers.

- Microcomputers: Personal computers such as desktops and laptops.

- Embedded computers: Built into devices such as washing machines, vehicles, and digital appliances.

According to purpose

- General-purpose computers: Perform many different tasks.

- Special-purpose computers: Designed for a specific task, such as controlling traffic lights.

According to data handled

- Analog computers: Process continuously varying physical quantities.

- Digital computers: Process discrete data, usually represented in binary form.

- Hybrid computers: Combine analog and digital processing.

TOPIC 2: COMPUTER SYSTEMS

2.1 Meaning of a computer system

A computer system consists of hardware, software, data, people, and procedures working together to process data into information.

2.2 Components of a computer system

1. Hardware: Physical parts of a computer.

2. Software: Programs and instructions that direct the computer.

3. Data: Raw facts entered into the computer.

4. Peopleware: People who use, operate, maintain, or manage computer systems.

5. Procedures: Rules and steps followed when operating the system.

2.3 Data and information

Data refers to raw facts that have not yet been processed into a useful form.

Information is processed data that is meaningful and useful.

Example:

- Data: 60, 75, 85

- Information: The average mark is \(73.33\).

2.4 Information processing cycle

The information processing cycle consists of:

- Collection of data

- Input of data

- Processing

- Output of information

- Storage

- Distribution or communication of information

2.5 Qualities of good information

Good information should be:

- Accurate

- Relevant

- Complete

- Timely

- Reliable

- Understandable

- Accessible to authorised users

- Cost-effective to obtain

TOPIC 3: COMPUTER HARDWARE

3.1 Definition

Computer hardware refers to the physical components of a computer system that can be seen and touched.

3.2 Input devices

Input devices enter data and instructions into a computer.



Keyboard and mouse

The keyboard enters text, numbers, and commands. The mouse controls the pointer and selects items on the screen.



Scanner

Converts paper documents and photographs into digital images.

Webcam

Captures video and images for video conferencing and recording.



Other input devices include:

- Microphone
- Touchscreen
- Barcode reader
- Joystick
- Graphics tablet
- Biometric scanner
- Optical Mark Recognition (OMR) reader

- Optical Character Recognition (OCR) reader

- Magnetic Ink Character Recognition (MICR) reader

3.3 Output devices

Output devices present processed information to users.



Monitor

Displays text, images, video, and other visual information.

Printer

Produces hard copies of documents and images.



Speakers

Produce sound output.

Other output devices include projectors, headphones, and plotters.

3.4 Central Processing Unit (CPU)

The CPU is the main processing component of a computer. It executes instructions and controls computer operations.

Its major components are:

- Arithmetic Logic Unit (ALU): Performs arithmetic calculations and logical comparisons.

- Control Unit (CU): Directs and coordinates computer operations.

- Registers: Very small, fast storage locations inside the CPU.

- Cache: High-speed memory that keeps frequently needed data and instructions close to the CPU.

3.5 Storage devices

Storage devices retain data and programs.

Device	Description
Hard Disk Drive (HDD)	Uses magnetic disks to store data.
Solid State Drive (SSD)	Uses flash memory and has no moving disk parts.
USB flash drive	Portable storage connected through a USB port.
Memory card	Small removable storage used in phones and cameras.
Optical disc	CD, DVD, or Blu-ray disc read using laser technology.
External drive	Storage connected externally to a computer.

3.6 Primary and secondary storage

Primary memory is used directly by the CPU during processing. It includes RAM and ROM.

- RAM: Temporary, volatile memory. Its contents are generally lost when power is removed.

- ROM: Non-volatile memory used to retain firmware or other essential instructions.

Secondary storage retains data for long-term use, such as SSDs, HDDs, and USB drives.

TOPIC 4: COMPUTER SOFTWARE

4.1 Definition

Software is a collection of programs, instructions, and related data that tells a computer what to do.

4.2 Types of software

A. System software

System software manages computer hardware and provides a platform for applications.

Examples:

- Windows

- Linux

- macOS

- Device drivers

- Utility programs

- Language translators

B. Application software

Application software helps users perform specific tasks.

Examples:

- Microsoft Word — document preparation

- Microsoft Excel — calculations and spreadsheets

- Microsoft PowerPoint — presentations

- Microsoft Access — database management

- Web browsers — accessing websites

- Accounting software — financial management

C. Utility software

Utility software helps maintain, protect, and optimise a computer.

Examples:

- Antivirus software

- Backup utilities

- File compression tools

- Disk management tools

- File recovery utilities

4.3 System software versus application software

System software	Application software

Manages computer resources	Helps users perform tasks
Provides a platform for applications	Runs on a suitable operating system
Often starts as part of system startup	Usually launched when required
Example: Windows	Example: Microsoft Word

4.4 Language translators

A language translator converts program instructions into a form the computer can execute.

- Compiler: Translates a program into machine code or another target form, commonly before execution.

- Interpreter: Executes program instructions through interpretation, often statement by statement.

- Assembler: Converts assembly language instructions into machine code.

TOPIC 5: OPERATING SYSTEMS

5.1 Definition

An operating system (OS) is system software that manages computer hardware and software resources and provides an interface through which users and applications interact with the computer.

Examples include Windows, Linux, Android, macOS, and iOS.

5.2 Functions of an operating system

1. Process management: Controls running programs and allocates CPU time.

2. Memory management: Allocates and releases main memory.

3. File management: Organises files and directories.

4. Device management: Coordinates hardware devices through drivers.

5. User interface: Provides graphical or command-based interaction.

6. Security management: Helps control access to system resources.

7. Error handling: Detects and responds to certain system errors.

8. Resource allocation: Distributes available resources among programs and users.

5.3 Types of operating systems

- Single-user: Designed primarily for one user at a time.

- Multi-user: Supports multiple users accessing system resources.

- Multitasking: Allows multiple tasks to run concurrently.

- Real-time: Designed to respond within specified timing constraints.

- Network operating system: Provides services and management for networked computers.

- Mobile operating system: Designed for mobile devices.

5.4 Graphical User Interface (GUI) and Command-Line Interface (CLI)

GUI	CLI
Uses windows, icons, menus, and pointers	Uses typed commands
Generally easier for beginners	Requires knowledge of commands
Uses a mouse or touch input frequently	Often uses a keyboard
Example: Windows desktop	Example: Command Prompt

5.5 Starting and shutting down a computer

Starting up (booting) is the process of loading the operating system into memory.

- Cold boot: Starting a computer from a powered-off state.

- Warm boot: Restarting a computer without completely switching off its power.

Proper shutdown steps:

1. Save open work.

2. Close applications.

3. Open the operating system's power menu.

4. Select Shut down.

5. Wait until the computer has powered off before disconnecting power, where applicable.

TOPIC 6: DESKTOP AND FILE MANAGEMENT

6.1 Desktop

The desktop is the main working area displayed after a graphical operating system loads.

Common desktop elements include:

- Icons and shortcuts

- Taskbar

- Start menu

- Notification area

- Desktop background

- Open application windows

6.2 Files and folders

A file is a named collection of data stored on a computer.

A folder is a container used to organise files and other folders.

Examples of file extensions:

Extension	File type
.docx	Word document
.xlsx	Excel workbook
.pptx	PowerPoint presentation
.pdf	Portable Document Format
.txt	Plain text
.jpg / .png	Image
.mp3	Audio
.mp4	Video
.exe	Windows executable program

6.3 Common file operations

- Create: Make a new file or folder.

- Open: Access an existing file.

- Save: Store changes to a file.

- Save As: Save a file under a different name, location, or format.

- Copy: Create another instance of a file.

- Move: Transfer a file to another location.

- Rename: Change the name of a file or folder.

- Delete: Remove a file or folder.

- Restore: Recover an item from a recycle or trash area, if available.

6.4 Common keyboard shortcuts

Shortcut	Function

Ctrl + C Copy

Ctrl + X Cut

Ctrl + V Paste

Ctrl + S Save

Ctrl + P Print

Ctrl + Z Undo

Ctrl + A Select all

Ctrl + F Find

Alt + Tab Switch between open applications

Win + E Open File Explorer

Win + D Show or hide the desktop

F2 Rename a selected item in File Explorer

TOPIC 7: COMPUTER CARE, SAFETY AND SECURITY

7.1 Computer care and maintenance

Computer maintenance involves activities that keep a computer working correctly and reliably.

Preventive maintenance

- Keep the computer clean and dust-free.

- Ensure adequate ventilation.

- Use a suitable surge protector or UPS where appropriate.

- Keep software updated.

- Back up important files.

- Scan for malware.

- Handle cables and peripherals carefully.

- Shut down the computer properly.

Corrective maintenance

Corrective maintenance involves diagnosing and repairing faults after they occur, such as replacing a damaged keyboard or repairing a faulty storage device.

7.2 Computer security threats

- Virus: Malicious code that attaches to files or programs and spreads when they are executed.

- Worm: Malware that can spread between systems, often through networks.

- Trojan horse: Malicious software disguised as legitimate software.

- Spyware: Software that secretly gathers information.

- Ransomware: Malware that locks or encrypts data and demands payment.

- Phishing: Fraudulent messages or websites designed to steal information.

- Unauthorised access: Access to a system or data without permission.

7.3 Measures for protecting computers

1. Use strong, unique passwords.

2. Enable multi-factor authentication where available.

3. Install reputable security software.

4. Keep the operating system and applications updated.

5. Avoid suspicious links and attachments.

6. Download software from trusted sources.

7. Back up important information regularly.

8. Lock the computer when leaving it unattended.

9. Restrict access to authorised users.

10. Avoid sharing confidential information on untrusted networks.

7.4 Health and safety when using computers

- Sit with your back supported.

- Position the monitor at a comfortable viewing height.

- Keep wrists in a neutral position.

- Take regular breaks to reduce eye strain.

- Provide sufficient lighting and ventilation.

- Keep liquids away from electrical equipment.

- Avoid overloaded electrical sockets.

- Use proper lifting techniques when moving heavy equipment.

TOPIC 8: BASIC WORD PROCESSING

8.1 Definition

Word processing is the use of computer software to create, edit, format, save, and print text documents.

Examples of word-processing applications include Microsoft Word and LibreOffice Writer.

8.2 Common word-processing operations

- Creating and saving documents

- Typing and editing text

- Selecting, copying, cutting, and pasting text

- Formatting fonts and paragraphs

- Setting page margins and orientation

- Inserting tables, images, and page numbers

- Checking spelling and grammar

- Printing documents

8.3 Text formatting

Common formatting options include:

- Bold: Makes text darker or thicker.

- Italic: Slants text.

- Underline: Places a line below text.

- Font size: Changes text size.

- Font colour: Changes text colour.

- Alignment: Left, centre, right, or justified.

- Line spacing: Controls vertical space between lines.

- Bullets and numbering: Organise information into lists.

8.4 Page layout

Page layout controls how a document is arranged on a page.

Important features include:

- Margins

- Page size

- Portrait or landscape orientation

- Page breaks

- Headers and footers

- Page numbering

- Columns

TOPIC 9: BASIC SPREADSHEET APPLICATIONS

9.1 Definition

A spreadsheet is an application that organises data in rows and columns and supports calculations, analysis, and chart creation.

Examples include Microsoft Excel and LibreOffice Calc.

9.2 Spreadsheet terminology

- Workbook: A spreadsheet file.

- Worksheet: A sheet inside a workbook.

- Row: A horizontal series of cells, identified by numbers.

- Column: A vertical series of cells, identified by letters.

- Cell: The intersection of a row and a column.

- Cell address: The location of a cell, such as A1.

- Formula: An expression used to calculate a result.

- Function: A predefined formula, such as SUM().

9.3 Common spreadsheet formulas

Formula	Purpose
=SUM(B2:B6)	Adds values
=AVERAGE(B2:B6)	Calculates the mean
=MAX(B2:B6)	Finds the largest value
=MIN(B2:B6)	Finds the smallest value
=COUNT(B2:B6)	Counts cells containing numbers
=IF(B2>=50,"Pass","Fail")	Returns a result based on a condition

Example: If a student scores 60, 70, and 80 in cells B2, C2, and D2, the average formula is:

=AVERAGE(B2:D2)

The result is 70.

TOPIC 10: BASIC DATABASE APPLICATIONS

10.1 Definition

A database is an organised collection of related data stored so that it can be accessed, managed, updated, and retrieved efficiently.

A Database Management System (DBMS) is software used to create and manage databases.

Examples include Microsoft Access and PostgreSQL.

10.2 Database terminology

- Table: Organises related data into rows and columns.

- Field: A column representing a particular attribute.

- Record: A row containing details about one item or person.

- Primary key: A field or combination of fields that uniquely identifies a record.

- Query: A request to retrieve or manipulate data.

- Form: An interface for entering or viewing data.

- Report: A formatted presentation of database information.

10.3 Uses of databases

- Student registration

- Employee records

- Hospital patient records

- Stock and inventory management

- Banking records

- Library management

10.4 Advantages of databases

- Reduce unnecessary data duplication.

- Improve data retrieval.

- Support controlled access to information.

- Improve consistency when designed properly.

- Support backup and recovery features.

- Allow reports and queries to be generated.

TOPIC 11: BASIC PRESENTATION APPLICATIONS

11.1 Definition

Presentation software is used to create and display information in a sequence of slides.

Examples include Microsoft PowerPoint and LibreOffice Impress.

11.2 Components of a presentation

- Slides

- Titles and text

- Images and diagrams

- Charts and tables

- Audio and video

- Themes and layouts

- Transitions and animations

- Speaker notes

11.3 Guidelines for effective presentations

1. Use a clear and consistent design.

2. Use readable font sizes.

3. Avoid overcrowding slides with text.

4. Use relevant images and diagrams.

5. Choose colours with sufficient contrast.

6. Use animations sparingly.

7. Proofread the content.

8. Practise before presenting.

TOPIC 12: COMPUTER NETWORKS AND THE INTERNET

12.1 Definition of a computer network

A computer network is a group of interconnected computers and devices that communicate and share resources.

12.2 Types of networks

- LAN (Local Area Network): Covers a limited area, such as a classroom, office, or college.

- MAN (Metropolitan Area Network): Covers a town or metropolitan area.

- WAN (Wide Area Network): Covers a large geographical area.

- WLAN (Wireless Local Area Network): A LAN that uses wireless communication, such as Wi-Fi.

12.3 Network devices

- Router: Directs data between networks.

- Switch: Connects devices within a LAN and forwards frames to appropriate ports.

- Modem: Converts signals to support communication over a particular transmission link.

- Wireless access point: Allows wireless devices to connect to a network.

- Network Interface Card (NIC): Enables a device to connect to a network.

12.4 Internet terminology

- Internet: A global network of interconnected networks.

- World Wide Web: A collection of interlinked resources accessed over the internet.

- Web browser: Software used to access websites.

- Search engine: A service used to find online information.

- URL: The address of a web resource.

- Email: Electronic messaging between users.

- Cloud computing: Accessing computing resources and services over a network.

12.5 Benefits of the internet

- Online learning
- Communication
- Research
- E-commerce
- Online banking
- Entertainment
- Sharing information
- Accessing government services

END-OF-MODULE 1 REVISION QUESTIONS

Section A: Short-answer questions

1. Define a computer.
2. State four characteristics of a computer.
3. Differentiate between data and information.
4. List five components of a computer system.
5. State four applications of computers.
6. Differentiate between hardware and software.
7. Name four input devices and four output devices.
8. State three functions of the CPU.
9. Differentiate between RAM and ROM.
10. Explain the difference between system software and application software.
11. State five functions of an operating system.
12. Differentiate between GUI and CLI.
13. Explain the difference between a file and a folder.
14. State five computer security threats.
15. Give five measures for protecting computer systems.
16. Define word processing and give two examples of word processors.
17. Explain the terms workbook, worksheet, cell, and formula.

18. Define a database and explain the purpose of a primary key.

19. Differentiate between LAN and WAN.

20. State four benefits of using the internet.

Section B: Structured questions

Question 1: Computer systems

a. Explain the four basic functions of a computer. (8 marks)

b. Describe four characteristics of a computer. (8 marks)

c. Explain four limitations of computers. (4 marks)

Question 2: Hardware and software

a. Describe the functions of the ALU and Control Unit. (4 marks)

b. Distinguish between primary and secondary storage. (4 marks)

c. Explain three types of application software, giving one example of each. (6 marks)

Question 3: Operating systems and security

a. Explain five functions of an operating system. (10 marks)

b. Describe four ways of protecting a computer against security threats. (8 marks)

c. State two safe practices when shutting down a computer. (2 marks)

Question 4: Office applications

a. Explain five functions of word-processing software. (5 marks)

b. Write a spreadsheet formula to calculate the average of cells B2 to B6. (2 marks)

c. Explain the difference between a database table, field, and record. (6 marks)

Question 5: Networks and the internet

a. Explain three types of computer networks. (6 marks)

b. State four uses of the internet. (4 marks)

c. Describe the functions of a router and a switch. (4 marks)

Quick revision checklist

Your revision progress

0 of 13

[ ] Computer definitions and functions

[ ] Computer system components

[ ] Input, output and storage devices

[ ] CPU and primary memory

[ ] System and application software

[ ] Operating system functions

[ ] File and folder management

[ ] Computer security and maintenance

[ ] Word processing

[ ] Spreadsheets and formulas

[ ] Database terminology

[ ] Presentation software

[ ] Computer networks and internet